

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – CLASS TEST (5 Questions)

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Precision (BL6)	Assessment:	Assessment Tool 1 – Class Test
Weightage:	40% of CIA	Total Marks:	50 (5 Questions × 10 Marks)
CO5 Statement:	<i>Create effective and interactive visualizations by integrating data wrangling and advanced visualization tools (BL6)</i>		
Student Name:	G Kushwanth	Reg. No.:	192424150
Section / Batch:	AI&DS	Date:	

Code of Conduct

I G Kushwanth(192424150) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G Kushwanth

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Uncertainty & Confidence Intervals	Uncertainty of point estimates, confidence intervals, trend estimation	1	10
Handling Overplotting	Overlapping points, jittering, transparency	2	10
Principles of Effective Visualization	Proportional ink principle, avoiding misleading axes	3	10
Accessibility & Misuse of Color	Color accessibility, avoiding misuse of color	4	10
Probability & Hypothetical Outcome Plots	Probability distribution and hypothetical outcome plots	5	10
GRAND TOTAL:		50 Marks	

Annexure A – Class Test

1. A survey estimates the average monthly income (₹) of two cities:

City	Mean Income	Confidence Interval
A	25,000	± 2,000
B	27,000	± 5,000

Represent this dataset using a suitable graph to visualize uncertainty. Justify your choice and interpret which city has more reliable estimates.

Answer:

Chart choice: A point-range chart (also called a bar-and-error-bar or dot-and-whisker plot) is best suited here. Each city is plotted as a single point (or bar) at its mean income, with a vertical line extending above and below it to the upper and lower bounds of the confidence interval. A box plot is a reasonable alternative if the underlying raw data were available, but since we are only given a summary mean and margin of error, the point-range plot is the more direct and honest representation.

Why this chart: A confidence interval expresses the range within which the true population mean is likely to fall, typically written as mean plus or minus a margin of error ($CI = \bar{x} \pm z \times SE$, where SE is the standard error of the estimate). Showing only the mean as a single bar would hide this uncertainty and could make two different estimates look equally trustworthy even when one is far less precise. Error bars make the width of uncertainty visually explicit, so the viewer can judge precision at a glance rather than trusting a single number.

Interpretation: City A has a mean income of Rs 25,000 with a narrow interval of $\pm$ Rs 2,000 (plausible range: Rs 23,000 to Rs 27,000). City B has a higher mean of Rs 27,000 but a much wider interval of $\pm$ Rs 5,000 (plausible range: Rs 22,000 to Rs 32,000). City A's interval is narrower relative to its mean, so its estimate is more precise and more reliable, even though its point estimate is lower than City B's. Notably, the two intervals overlap considerably (City A's upper bound of Rs 27,000 coincides with City B's mean, and City B's lower bound of Rs 22,000 falls below City A's own mean), which means we cannot confidently say the two cities' true average incomes are actually different from each other at this confidence level. A larger sample size for City B would likely narrow its interval and allow a more conclusive comparison.

2. A dataset contains exam scores of 200 students, where many students have similar scores:

Score	Frequency
50	30
60	45
70	60
80	40
90	25

Choose an appropriate visualization to avoid overplotting. Explain how techniques like jittering or transparency can improve clarity.

Answer:

The problem: Plotting all 200 individual student scores as a raw scatter or one-dimensional dot plot along a single axis will cause heavy overplotting, because many students share the exact same score (for example, 60 different students all scored 70). Identical values land exactly on top of one another, so a plain dot plot would only ever show five visible dots (one per unique score) no matter how many students actually got that score, completely hiding the frequency information the question is asking us to visualize.

Chart choice: Since the data is already aggregated into five score bins with counts, the cleanest solution is a histogram or bar chart of frequency versus score, which needs no overplotting fix at all because each bar's height already encodes the count directly. If instead we had the 200 raw, unaggregated data points and wanted to show individual students (for example on a strip plot along one axis, or a scatter plot against a second variable such as study hours), a jittered strip plot combined with transparency is the standard remedy.

How jittering helps: Jittering adds a small random offset (typically a fraction of the spacing between categories, e.g. plus or minus 0.1 to 0.2 units) to each point's position, perpendicular to the value being measured. Points that share the same score no longer sit exactly on top of one another; instead they spread into a visible cloud whose width and density reflect how many students share that score, so a score of 70 with 60 students produces a noticeably wider, denser cloud than a score of 90 with only 25 students.

How transparency helps: Setting each point's opacity to a low value (commonly alpha 0.3 to 0.5) means a single point looks pale, but wherever many points overlap the colors stack and the region appears progressively darker. This lets the eye read relative density directly from how dark or saturated a region looks, without needing jitter to physically separate every point. Used together, jittering separates points spatially and transparency encodes density through opacity; for very large datasets, a 2D histogram, hexbin plot, or kernel density estimate (KDE) plot are further alternatives that bin or smooth the data instead of showing individual points at all.

3. A company reports revenue growth:

Year	Revenue (₹ lakhs)
2021	100
2022	110
2023	115

Create a graph that avoids misleading axes and follows the proportional ink principle. Explain how improper axis scaling could distort interpretation.

Answer:

Chart choice: A simple bar chart with the y-axis (Revenue) starting at zero, or alternatively a line chart with a zero baseline, best follows the proportional ink principle: the amount of visual ink (bar height or area) used to represent each value should be proportional to the actual data value. With a zero baseline, revenue of 100, 110, and 115 lakhs for 2021 to 2023 produces bars in the height ratio 100:110:115, which exactly matches the ratio of the underlying numbers.

Quantifying the growth: Year-on-year growth is $(110-100)/100 = 10$ percent from 2021 to 2022, and $(115-110)/110$ is approximately 4.5 percent from 2022 to 2023. The compound annual growth rate (CAGR) across the full period is $(115/100)^{1/2} - 1$, which is approximately 7.2 percent per year. These are genuinely modest, single-digit growth rates, and an honest chart should visually communicate that the company is growing steadily but not dramatically.

How truncation distorts: If the y-axis were truncated to start at, say, 95 lakhs instead of zero, the visible bar heights would represent only the differences above 95 (5, 15, and 20 units respectively), so the 2023 bar would appear four times taller than the 2021 bar even though actual revenue only grew by 15 percent overall. This inflates what statisticians call the "lie factor" of the chart, defined as the size of the effect shown in the graphic divided by the size of the effect in the underlying data; a lie factor far from 1 signals a misleading visualization. Truncated axes are one of the most common ways charts mislead viewers who judge trends by eye rather than by reading the axis labels carefully, so keeping a zero baseline (or clearly annotating any deliberate truncation) is essential for an honest revenue chart.

4. A dataset shows profit/loss across departments:

Department	Status
HR	Profit
Sales	Loss
IT	Profit
Finance	Loss

Design a suitable visualization ensuring accessibility. Suggest color choices and explain how to avoid misuse of color for colorblind users.

Answer:

Chart choice: A diverging bar chart works well here, with each department's bar extending from a central zero line: Profit departments (HR, IT) extend to one side and Loss departments (Sales, Finance) extend to the other. This encodes the sign of the value through direction and position on the axis, not through color alone, so the meaning survives even if color is removed entirely.

Color choice: Roughly 8 percent of men and 0.5 percent of women have some form of red-green color vision deficiency (protanopia or deuteranopia), which makes the traditional red-for-loss, green-for-profit pairing a poor choice since both colors can appear as similar muddy tones to these viewers. A safer choice is a colorblind-friendly pair from the Okabe-Ito palette, for example blue (hex #0072B2) for Profit and vermilion-orange (hex #D55E00) for Loss; these two hues remain clearly distinguishable under protanopia, deuteranopia, and the much rarer blue-yellow deficiency (tritanopia). Tools such as ColorBrewer2.org or the viridis palette family are good references for pre-tested, perceptually uniform, colorblind-safe color schemes.

Avoiding misuse of color: Color should never be the only channel carrying essential information. Redundant encoding should be layered on top of color: label each bar directly with the department name and the word "Profit" or "Loss"; use different fill patterns such as solid versus diagonal hatching so bars remain distinguishable in grayscale printing; and ensure sufficient contrast (a contrast ratio of at least 3:1 against the background, per WCAG accessibility guidelines) so low-vision users can still read the chart. Together, direction, labels, pattern, and a colorblind-safe palette make the visualization accessible to the widest possible audience rather than relying on color perception alone.

5. A factory produces light bulbs with a 5% defect rate. A quality control engineer wants to visualize the probability of defects in batches of 20 bulbs.

Represent the probability distribution using a suitable graph (e.g., hypothetical outcome plot or probability plot). Interpret the likelihood of observing 0, 1, or more defective bulbs.

Answer:

Setting up the model: Each bulb is either defective or not, independently, with a fixed defect probability of $p = 0.05$, and we are examining a fixed batch size of $n = 20$ bulbs. This is exactly the setup for a Binomial distribution, with probability mass function $P(X = k) = C(n, k) \times p^k \times (1-p)^{(n-k)}$, where $C(n, k)$ is the number of ways to choose k defective bulbs out of 20 and X is the number of defective bulbs in a batch.

Chart choice: The probability distribution can be represented as a bar chart of the probability mass function, with the number of defective bulbs (0, 1, 2, 3, ...) on the x-axis and the probability of that exact count on the y-axis; each bar's height sums to 1 across all possible outcomes. A complementary, more intuitive alternative is a hypothetical outcome plot (HOP): instead of one static bar chart, the display animates through many individually simulated batches of 20 bulbs, showing a sequence of frames where each frame is one plausible outcome drawn from the distribution. Watching many such frames helps viewers develop an intuitive feel for how often each outcome actually occurs, which some people find easier to grasp than reading probability values off a bar chart.

Calculating the probabilities: $P(0 \text{ defects}) = C(20, 0) \times 0.05^0 \times 0.95^{20} = 0.95^{20}$, which is approximately 0.359 (about 35.8 percent). $P(1 \text{ defect}) = C(20, 1) \times 0.05^1 \times 0.95^{19} = 20 \times 0.05 \times 0.95^{19}$, which is approximately 0.377 (about 37.7 percent). $P(2 \text{ defects}) = C(20, 2) \times 0.05^2 \times 0.95^{18} = 190 \times 0.0025 \times 0.95^{18}$, which is approximately 0.189 (about 18.9 percent). Summing these, $P(0, 1, \text{ or } 2 \text{ defects})$ is approximately 92.4 percent, so $P(3 \text{ or more defects})$ is only about 7.6 percent, and $P(2 \text{ or more defects})$ is approximately 1 minus $P(0)$ minus $P(1)$, or about 26.4 percent.

Interpretation: The expected (mean) number of defective bulbs per batch is $E(X) = n \times p = 20 \times 0.05 = 1$, with variance $n \times p \times (1-p) = 20 \times 0.05 \times 0.95 = 0.95$ and a standard deviation of about 0.97 bulbs, confirming that "around 1 defective bulb per batch" is indeed the typical outcome. In practical terms: it is quite likely (about 36 percent of batches) that a batch of 20 bulbs has zero defects, almost equally likely (about 38 percent) to have exactly one defective bulb, and there is still a meaningful chance (about 26 percent) of seeing two or more defective bulbs in a batch, though seeing three or more is comparatively rare (under 8 percent). This kind of visualization helps a quality control engineer set realistic acceptance thresholds, for example flagging a batch for further inspection only if it has 3 or more defects, since that outcome is statistically unusual under normal 5 percent defect-rate conditions.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions		
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation		
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis		
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers		
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____